sampling, as commonly used for open-ended text generation (Radford et al., 2019; Brown et al., 2020; Thoppilan et al., 2022), to achieve this goal. One should note that self-consistency can be applied only to problems where the final answer is from a fixed answer set, but in principle this approach can be extended to open-text generation problems if a good metric of consistency can be defined between multiple generations, e.g., whether two answers agree or contradict each other.

3 EXPERIMENTS

We conducted a series of experiments to compare the proposed self-consistency method with existing approaches on a range of reasoning benchmarks. We find that self-consistency robustly improves reasoning accuracy for every language model considered, spanning a wide range of model scales.

3.1 EXPERIMENT SETUP

Tasks and datasets. We evaluate self-consistency on the following reasoning benchmarks.[3]

- **Arithmetic reasoning**. For these tasks, we used the Math Word Problem Repository (Koncel-Kedziorski et al., 2016), including AddSub (Hosseini et al., 2014), MultiArith (Roy & Roth, 2015), and ASDiv (Miao et al., 2020). We also included AQUA-RAT (Ling et al., 2017), a recently published benchmark of grade-school-math problems (GSM8K; Cobbe et al., 2021), and a challenge dataset over math word problems (SVAMP; Patel et al., 2021).
- **Commonsense reasoning**. For these tasks, we used CommonsenseQA (Talmor et al., 2019), StrategyQA (Geva et al., 2021), and the AI2 Reasoning Challenge (ARC) (Clark et al., 2018).
- **Symbolic Reasoning**. We evaluate two symbolic reasoning tasks: last letter concatenation (e.g., the input is "Elon Musk" and the output should be "nk"), and Coinflip (e.g., a coin is heads-up, after a few flips is the coin still heads-up?) from Wei et al. (2022).

Language models and prompts. We evaluate self-consistency over four transformer-based language models with varying scales:

- UL2 (Tay et al., 2022) is an encoder-decoder model trained on a mixture of denoisers with 20-billion parameters. UL2 is completely open-sourced[4] and has similar or better performance than GPT-3 on zero-shot SuperGLUE, with only 20B parameters and thus is more compute-friendly;
- GPT-3 (Brown et al., 2020) with 175-billion parameters. We use two public engines *code-davinci-001* and *code-davinci-002* from the Codex series (Chen et al., 2021) to aid reproducibility;[5]
- LaMDA-137B (Thoppilan et al., 2022) is a dense left-to-right, decoder-only language model with 137-billion parameters, pre-trained on a mixture of web documents, dialog data and Wikipedia;
- PaLM-540B (Chowdhery et al., 2022) is a dense left-to-right, decoder-only language model with 540-billion parameters, pre-trained on a high quality corpus of 780 billion tokens with filtered webpages, books, Wikipedia, news articles, source code, and social media conversations.

We perform all experiments in the few-shot setting, without training or fine-tuning the language models. For a fair comparison we use the same prompts as in Wei et al. (2022): for all arithmetic reasoning tasks we use the same set of 8 manually written exemplars; for each commonsense reasoning task, 4-7 exemplars are randomly chosen from the training set with manually composed chain-of-thought prompts.[6] Full details on the prompts used are given in Appendix A.3.

Sampling scheme. To sample diverse reasoning paths, we followed similar settings to those suggested in Radford et al. (2019); Holtzman et al. (2020) for open-text generation. In particular, for UL2-20B and LaMDA-137B we applied temperature sampling with $T = 0.5$ and truncated at the top-k ($k = 40$) tokens with the highest probability, for PaLM-540B we applied $T = 0.7, k = 40$, and for GPT-3 we use $T = 0.7$ without top-k truncation. We provide an ablation study in Section 3.5 to show that self-consistency is generally robust to sampling strategies and parameters.

[3] By default we use the test split for all datasets if the labels are available for evaluation. For CommonsenseQA we use the dev split; for StrategyQA we use the question-only set from BIG-bench collaboration (2021): https://github.com/google/BIG-bench/tree/main/bigbench/benchmark_tasks/strategyqa.

[4] Model checkpoints at https://github.com/google-research/google-research/tree/master/ul2.

[5] Public API available at https://openai.com/api/.

[6] Self-consistency is robust to different sets of prompts and we provide a study in Appendix A.1.2.

3.2 MAIN RESULTS

We report the results of self-consistency averaged over 10 runs, where we sampled 40 outputs independently from the decoder in each run. The baseline we compare to is chain-of-thought prompting with greedy decoding (Wei et al., 2022), referred to as **CoT-prompting**, which has been previously used for decoding in large language models (Chowdhery et al., 2022).

Arithmetic Reasoning The results are shown in Table 2.[7] Self-consistency improves the arithmetic reasoning performance over **all four language models** significantly over chain-of-thought prompting. More surprisingly, the gains become more significant when the language model's scale increases, e.g., we see +3%-6% absolute accuracy improvement over UL2-20B but +9%-23% for LaMDA-137B and GPT-3. For larger models that already achieve high accuracy on most tasks (e.g., GPT-3 and PaLM-540B), self-consistency still contributes significant additional gains with +12%-18% absolute accuracy on tasks like AQuA and GSM8K, and +7%-11% on SVAMP and ASDiv. With self-consistency, we achieve new state-of-the-art results on almost all tasks: despite the fact that self-consistency is unsupervised and task-agnostic, these results compare favorably to existing approaches that require task-specific training, or fine-tuning with thousands of examples (e.g., on GSM8K).

	Method	AddSub	MultiArith	ASDiv	AQuA	SVAMP	GSM8K
	Previous SoTA	**94.9**^{a}	60.5^a	75.3^b	37.9^c	57.4^d	35^e / 55^g
UL2-20B	CoT-prompting	18.2	10.7	16.9	23.6	12.6	4.1
	Self-consistency	24.8 (+6.6)	15.0 (+4.3)	21.5 (+4.6)	26.9 (+3.3)	19.4 (+6.8)	7.3 (+3.2)
LaMDA-137B	CoT-prompting	52.9	51.8	49.0	17.7	38.9	17.1
	Self-consistency	63.5 (+10.6)	75.7 (+23.9)	58.2 (+9.2)	26.8 (+9.1)	53.3 (+14.4)	27.7 (+10.6)
PaLM-540B	CoT-prompting	91.9	94.7	74.0	35.8	79.0	56.5
	Self-consistency	93.7 (+1.8)	99.3 (+4.6)	81.9 (+7.9)	48.3 (+12.5)	86.6 (+7.6)	74.4 (+17.9)
GPT-3 Code-davinci-001	CoT-prompting	57.2	59.5	52.7	18.9	39.8	14.6
	Self-consistency	67.8 (+10.6)	82.7 (+23.2)	61.9 (+9.2)	25.6 (+6.7)	54.5 (+14.7)	23.4 (+8.8)
GPT-3 Code-davinci-002	CoT-prompting	89.4	96.2	80.1	39.8	75.8	60.1
	Self-consistency	91.6 (+2.2)	**100.0** (+3.8)	**87.8** (+7.6)	**52.0** (+12.2)	**86.8** (+11.0)	**78.0** (+17.9)

Table 2: Arithmetic reasoning accuracy by self-consistency compared to chain-of-thought prompting (Wei et al., 2022). The previous SoTA baselines are obtained from: a: Relevance and LCA operation classifier (Roy & Roth, 2015), b: Lan et al. (2021), c: Amini et al. (2019), d: Pi et al. (2022), e: GPT-3 175B finetuned with 7.5k examples (Cobbe et al., 2021), g: GPT-3 175B finetuned plus an additional 175B verifier (Cobbe et al., 2021). The best performance for each task is shown in bold.

	Method	CSQA	StrategyQA	ARC-e	ARC-c	Letter (4)	Coinflip (4)
	Previous SoTA	**91.2**^{a}	73.9^b	86.4^c	75.0^c	N/A	N/A
UL2-20B	CoT-prompting	51.4	53.3	61.6	42.9	0.0	50.4
	Self-consistency	55.7 (+4.3)	54.9 (+1.6)	69.8 (+8.2)	49.5 (+6.8)	0.0 (+0.0)	50.5 (+0.1)
LaMDA-137B	CoT-prompting	57.9	65.4	75.3	55.1	8.2	72.4
	Self-consistency	63.1 (+5.2)	67.8 (+2.4)	79.3 (+4.0)	59.8 (+4.7)	8.2 (+0.0)	73.5 (+1.1)
PaLM-540B	CoT-prompting	79.0	75.3	95.3	85.2	65.8	88.2
	Self-consistency	80.7 (+1.7)	**81.6** (+6.3)	**96.4** (+1.1)	**88.7** (+3.5)	70.8 (+5.0)	91.2 (+3.0)
GPT-3 Code-davinci-001	CoT-prompting	46.6	56.7	63.1	43.1	7.8	71.4
	Self-consistency	54.9 (+8.3)	61.7 (+5.0)	72.1 (+9.0)	53.7 (+10.6)	10.0 (+2.2)	75.9 (+4.5)
GPT-3 Code-davinci-002	CoT-prompting	79.0	73.4	94.0	83.6	70.4	99.0
	Self-consistency	81.5 (+2.5)	79.8 (+6.4)	96.0 (+2.0)	87.5 (+3.9)	**73.4** (+3.0)	**99.5** (+0.5)

Table 3: Commonsense and symbolic reasoning accuracy by self-consistency compared to chain-of-thought prompting (Wei et al., 2022). The previous SoTA baselines are obtained from: a: DeBERTaV3-large + KEAR (Xu et al., 2021b), b: Chowdhery et al. (2022), c: UnifiedQA-FT (Khashabi et al., 2020). The best performance for each task is shown in bold.

[7] The standard deviation of self-consistency is ≤ 0.5 for all tasks and is thus omitted in the table. Please refer to Figure 2, Figure 7 and 8 for the standard deviations under varying numbers of sampled paths.

Commonsense and Symbolic Reasoning Table 3 shows the results on commonsense and symbolic reasoning tasks. Similarly, self-consistency yields large gains across all four language models, and obtained SoTA results on 5 out of 6 tasks. For symbolic reasoning, we test the out-of-distribution (OOD) setting where the input prompt contains examples of 2-letters or 2-flips but we test examples of 4-letters and 4-flips (this setting is more challenging as PaLM-540B or GPT-3 can already achieve perfect in-distribution accuracy). In this challenging OOD setting, the gain of self-consistency is still quite significant compared to CoT-prompting with sufficient model sizes.

To show the effect of the number of sampled reasoning paths, we plot the accuracy (mean and standard deviation over 10 runs) with respect to varying numbers of sampled paths (1, 5, 10, 20, 40) in Figure 2. The results show that sampling a higher number (e.g., 40) of reasoning paths leads to a consistently better performance, further emphasizing the importance of introducing diversity in the reasoning paths. In Table 4, we show self-consistency yields a richer set of reasoning paths compared to greedy decoding with a few example questions from two tasks.

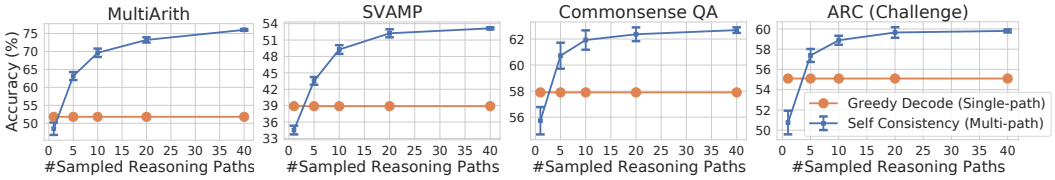

Figure 2: Self-consistency (blue) significantly improves accuracy over CoT-prompting with greedy decoding (orange) across arithmetic and commonsense reasoning tasks, over LaMDA-137B. Sampling a higher number of diverse reasoning paths consistently improves reasoning accuracy.

	GSM8K	StrategyQA
Question	*Henry made two stops during his 60-mile bike trip. He first stopped after 20 miles. His second stop was 15 miles before the end of the trip. How many miles did he travel between his first and second stops?*	*Yes or no: Is Albany, Georgia the most populous US Albany?*
Greedy Decode	He traveled 60 miles in total. He stopped after 20 miles, so he traveled 60 - 20 = 40 miles between the first and second stops. The answer is 40.	Albany, Georgia is the most populous US Albany. Thus, the answer is yes.
Sampled Path 1	Henry travelled a total of 60 miles. His first stop was 20 miles in, and his second stop was 15 miles before the end. So between his first and second stops he travelled 60 - 20 -15 = 25 miles. The answer is 25.	The most populous US Albany is Albany, New York. Thus, Albany, Georgia is not the most populous US Albany. So the answer is no.
Sampled Path 2	He made two stops during a 60-mile trip. The first was 20 miles into the trip. The second was 15 miles before the end of the trip. This means the second stop was 60 - 15 = 45 miles into the trip. Since he made the stops in order, the second stop must have been 45 - 20 = 25 miles after the first stop. The answer is 25.	Albany, Georgia has a population of about 88,000. Albany, New York has a population of about 95,000. Thus, Albany, Georgia is not the most populous US Albany. So the answer is no.

Table 4: Examples where self-consistency helps repair the errors over greedy decode, on PaLM-540B. Two sampled reasoning paths that are consistent with the ground truth are shown.

3.3 Self-Consistency Helps When Chain-of-Thought Hurts Performance

Ye & Durrett (2022) show that sometimes chain-of-thought prompting could hurt performance compared to standard prompting in few-shot in-context learning. Here we perform a study using self-consistency to see if it can help fill in the gap, over a set of common NLP tasks, including (1) Closed-Book Question Answering: BoolQ (Clark et al., 2019), HotpotQA (Yang et al., 2018), and (2) Natural Language Inference: e-SNLI (Camburu et al., 2018), ANLI (Nie et al., 2020) and RTE (Dagan et al., 2005; Bar-Haim et al., 2006; Giampiccolo et al., 2007; Bentivogli et al., 2009).

The results over PaLM-540B are shown in Table 5. For some tasks (e.g., ANLI-R1, e-SNLI, RTE), adding chain-of-thought does hurt performance compared to standard prompting (Brown et al., 2020), but self-consistency is able to robustly boost the performance and outperform standard prompting, making it a reliable way to add rationales in few-shot in-context learning for common NLP tasks.

	ANLI R1 / R2 / R3	e-SNLI	RTE	BoolQ	HotpotQA (EM/F1)
Standard-prompting (no-rationale)	69.1 / 55.8 / 55.8	85.8	84.8	71.3	27.1 / 36.8
CoT-prompting (Wei et al., 2022)	68.8 / 58.9 / 60.6	81.0	79.1	74.2	28.9 / 39.8
Self-consistency	**78.5 / 64.5 / 63.4**	**88.4**	**86.3**	**78.4**	**33.8 / 44.6**

Table 5: Compare Standard/CoT prompting with self-consistency on common NLP tasks.